AI2: Estimation beyond Maximum Likelihood

Bayesian, MAP and MLE Estimation

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Outline

By the end of this lecture you will be able to:

- Formally describe and understand the conceptual difference between the Bayesian and Frequentist frameworks
- Understand the wider context of Maximum Likelihood estimation
- Learn about Maximum A-Posteriori (MAP) and fully Bayesian estimation.

Supervised vs. Unsupervised Learning

- **Unsupervised Learning:** Fit a probabilistic model to input samples $\{x_1, \ldots, x_N\}$.
- **Supervised Learning:** Learn a mapping from inputs $\{x_1, \ldots, x_n\}$ to the corresponding targets $\{y_1, \ldots, y_n\}$.

Bayesian Estimation

Let D denote data, and θ the parameters of our model.

- Bayesians represent uncertainty by treating θ as a random variable with a prior distribution. Parameters and data are often continuous-valued; let $p(\theta)$ be the prior PDF.
- Return the posterior distribution of θ given data D:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$$

- Bayesian estimation does not operate through optimisation to obtain best parameter values.
- $p(D) = \int d\theta p(D|\theta)p(\theta)$ is called marginal likelihood, or evidence. Often intractable to compute! (Bayesians use either variational approximations, or sampling methods e.g. MCMC.)
- Predictive distributions for a new example:
  - in supervised learning: $p(Y|X, D) = \int d\theta \, p(Y|X, \theta)p(\theta|D)$
  - in unsupervised learning: $p(X|D) = \int d\theta \, p(X|\theta)p(\theta|D)$

Maximum A Posteriori Estimation

- Maximum A Posteriori (**MAP**): The most probable value of θ is where the posterior takes its maximum. Note: $p(D)$ has no influence on this. Hence, the MAP is much easier to compute than the full Bayesian estimate.

$$\begin{aligned}\theta_{MAP} &= \arg\max_{\theta}\ p(\theta|D) \\ &= \arg\max_{\theta}\ \{\log[p(D|\theta)] + \log[p(\theta)]\}\end{aligned}$$

- MAP use optimisation to find the parameter value that maximises the posterior probability.
- While it just returns a point-estimate, it is a conceptually Bayesian approach, as it incorporates a prior belief over parameter values.

Maximum Likelihood estimation - Frequentist Framework

- Maximum Likelihood (**ML**): Maximize likelihood (there is no prior):

$$\theta_{ML} = \arg\max_{\theta} \log[p(D|\theta)]$$

- ML can overfit with small datasets; MAP incorporates priors (i.e. pre-existing information) to mitigate this. Requires reliable prior info.
- Both ML and MAP use optimisation to find their best parameter values. ML has no notion of probability distribution over parameter values - all uncertainty in parameters are due to randomness in the data - the frequentist perspective.

ML is good when n is large

Imagine many data sets generated from some $P(\mathbf{x}_1, \ldots \mathbf{x}_n | \theta)$. From each we estimate θ. Denote $\hat{\theta}$ our estimate, and look at the expected square error from the true parameter value θ^*:

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \mathbb{E}[\hat{\theta}^2] - 2\mathbb{E}[\hat{\theta}]\theta^* + (\theta^*)^2$$

Add & subtract $\mathbb{E}[\hat{\theta}]^2$ to complete the square:

$$= \mathbb{E}[\hat{\theta}^2] + (\theta^* - \mathbb{E}[\hat{\theta}])^2 - \mathbb{E}[\hat{\theta}]^2$$

Rearrange (note that the cross-term $-2\mathbb{E}[\hat{\theta} \cdot \mathbb{E}[\hat{\theta}]] = 0$) to conclude:

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \underbrace{(\theta^* - \mathbb{E}[\hat{\theta}])^2}_{\mathrm{Bias}[\hat{\theta}]^2} + \underbrace{\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]}_{\mathrm{Var}[\hat{\theta}]}$$

The MLE is known to be *asymptotically unbiased* hence if we use ML, then the 1st term vanishes with large $n \to \infty$. In addition, $\mathrm{Var}(\hat{\theta})$ is of order $1/n$.

With finite n, the two terms are in a trade-off.

Summary

- Probabilities provide the key tool to handle uncertainty in AI
- Bayesian and Frequentist frameworks differ conceptually in the way they approach parameter estimation and uncertainty handling.
  - Bayesian Approaches integrate both aleatoric and epistemic uncertainty without distinction into a unified probabilistic framework by combining prior knowledge with observed data to produce a posterior distribution.
  - MAP estimation incorporates prior knowledge to model epistemic uncertainty, but simplifies the posterior distribution to a point estimate.
  - Maximum Likelihood Estimation is a frequentist method that does not directly model epistemic uncertainty, but can be complemented with post hoc methods like confidence intervals to do that.
- Maximum Likelihood (ML) works well when the sample size is large.
- Next week: Maximum Likelihood Estimation; Likelihood based learning.

Further readings

- Murphy, K. P. (2012). Machine Learning: A Probabilistic Perspective. MIT Press.

  Provides an excellent comparison of MLE and Bayesian methods, and their approaches to uncertainty.
- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.

  Explains Bayesian inference, MAP, and MLE in detail with applications in machine learning.
- Ghahramani, Z. (2015). Probabilistic machine learning and artificial intelligence. Nature, 521(7553), 452-459.

  Discusses probabilistic frameworks for handling uncertainty.
- Andrew Gelman et al. (2013). Bayesian Data Analysis. Chapman and Hall/CRC.

  A comprehensive resource for Bayesian methods and their philosophical differences from frequentist approaches.

Estimating the mean of a Bernoulli

Problem Setup

Suppose we flip a coin n times to find out its bias.

Bernoulli Distribution: $X \sim \text{Bernoulli}(\theta)$

- $X = 1$ with probability θ
- $X = 0$ with probability $1 - \theta$

Goal: Estimate θ (the mean of the Bernoulli, or the bias of the coin) from observed data. (A fair coin would have $\theta = 0.5$. A biased coin has a different θ.)

- Given n samples: $X_1, X_2, \ldots, X_n$
- Let $S = \sum_{i=1}^{n} X_i$ (number of successes)

Observed data: 0 1 1 1 0 1 1 0 1 1

$S = 7$, $n = 10$

Maximum Likelihood Estimation (MLE)

Step 1: Define the Likelihood Function

$$L(\theta) = \prod_{i=1}^{n} \theta^{X_i}(1-\theta)^{1-X_i}$$

Step 2: Compute the Log-Likelihood

$$\log L(\theta) = \sum_{i=1}^{n} X_i \log\theta + \sum_{i=1}^{n}(1-X_i)\log(1-\theta)$$

Step 3: Differentiate and Solve for θ

$$\frac{d}{d\theta}\log L(\theta) = \frac{S}{\theta} + \frac{n-S}{1-\theta}\cdot(-1) = 0$$

Step 4: Solve for MLE Estimate

$$\hat{\theta}_{\mathrm{MLE}} = \frac{S}{n}$$

Example: If $S = 7$, $n = 10$, then

$$\hat{\theta}_{\mathrm{MLE}} = \frac{7}{10} = 0.7$$

Maximum A Posteriori (MAP) Estimation

Step 1: Define a Prior Distribution

$$\theta \sim \mathsf{Beta}(\alpha, \beta)$$

Step 2: Compute the Posterior Distribution

$$p(\theta|X) \propto p(X|\theta)p(\theta)$$

Beta Distribution

Definition: The Beta distribution is defined as:

$$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

where $B(\alpha, \beta)$ is the Beta function:

$$B(\alpha, \beta) = \int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1} d\theta$$

Intuition: The Beta distribution is flexible and models beliefs about probabilities. Examples:

- $\alpha = \beta = 1$ (Uniform prior)
- $\alpha > \beta$ (More weight on higher values of θ)
- $\beta > \alpha$ (More weight on lower values of θ)

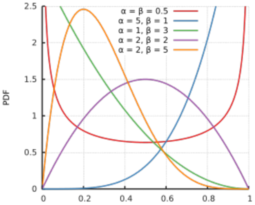

Figure: https://en.wikipedia.org/wiki/Beta_distribution

Maximum A Posteriori (MAP) Estimation

Step 1: Define a Prior Distribution

$$\theta \sim \mathrm{Beta}(\alpha, \beta)$$

Step 2: Compute the Posterior Distribution. For concreteness we take $\alpha = 2, \beta = 2$.

$$p(\theta|X) \propto p(X|\theta)p(\theta)$$

Using the likelihood function:

$$p(X|\theta) = \theta^S (1-\theta)^{n-S}$$

and the prior:

$$p(\theta) = \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

we get:

$$p(\theta|X) \propto \theta^{S+\alpha-1}(1-\theta)^{(n-S)+\beta-1}$$

Note, this matches the form of another Beta distribution:

$$\theta|X \sim \mathrm{Beta}(S+\alpha, n-S+\beta)$$

Step 3: Find the Maximum of the Posterior

$$\hat{\theta}_{\mathrm{MAP}} = \frac{S+\alpha-1}{n+\alpha+\beta-2} = \frac{S+1}{n+2} = \frac{8}{12} \approx 0.67$$

Preparation for Bayesian Estimation

Recall the Beta Distribution $\theta \sim B(\alpha, \beta)$

$$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

Its mean is (derivation on the next slide)

$$E[\theta] = \frac{\alpha}{\alpha + \beta}.$$

Recall that our posterior was also Beta, namely
$\theta|X \sim B(S + \alpha, n - S + \beta)$. Therefore, its mean will be...

$$E[\theta|X] = \frac{S + \alpha}{n + \alpha + \beta}.$$

Proof that the mean of $\theta \sim B(\alpha, \beta)$ is $E[\theta] = \frac{\alpha}{\alpha+\beta}$

Expectation Integral (by definition)

$$E[\theta] = \int_0^1 \theta p(\theta) d\theta$$

Substitute the PDF of Beta distribution

$$E[\theta] = \int_0^1 \theta \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)} d\theta$$

Factor Terms

$$E[\theta] = \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha}(1-\theta)^{\beta-1} d\theta$$

Recognize Beta Function Form The integral is the Beta function $B(\alpha+1, \beta)$, so:

$$E[\theta] = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)}$$

Use Beta Function Identity (proof on the next slide)

$$B(\alpha+1, \beta) = \frac{\alpha}{\alpha+\beta} B(\alpha, \beta)$$

Final Result:

$$E[\theta] = \frac{\alpha}{\alpha+\beta}$$

Proof of the Beta function identity $B(\alpha+1,\beta) = \frac{\alpha}{\alpha+\beta}B(\alpha,\beta)$

Recall the definition of the Beta Function

$$B(\alpha+1,\beta) = \int_0^1 \theta^{\alpha}(1-\theta)^{\beta-1}d\theta$$

Integration by Parts

- Let $u = \theta^{\alpha}$, so $du = \alpha\theta^{\alpha-1}d\theta$.
- Let $dv = (1-\theta)^{\beta-1}d\theta$, so $v = \frac{-(1-\theta)^{\beta}}{\beta}$.

Applying integration by parts:

$$B(\alpha+1,\beta) = \frac{\alpha}{\alpha+\beta}B(\alpha,\beta)$$

Now we are ready to give the Bayesian estimate.

Bayesian Estimation (Posterior Mean)

Goal: Compute Expected Posterior Value

$$E[\theta|X] = \int_0^1 \theta p(\theta|X) d\theta$$

Since the posterior is Beta-distributed as $\theta|X \sim B(S+\alpha, n-S+\beta)$,

$$E[\theta|X] = \frac{S+\alpha}{n+\alpha+\beta}.$$

Example Calculation when the prior parameters are $\alpha = 2, \beta = 2$:

$$E[\theta|X] = \frac{7+2}{10+2+2} = \frac{9}{14} \approx 0.64$$

Comparison of Methods

Taking $\alpha = 2, \beta = 2$:

Method	Formula	Estimate (S=7, n=10)
MLE	S/n	0.7
MAP	$\frac{S+\alpha-1}{n+\alpha+\beta-2}$	0.67
Bayesian	$\frac{S+\alpha}{n+\alpha+\beta}$	0.64

Note: Different choices of the prior parameters (α, β) would lead to different estimates.

Here we saw an example of the work involved in each of the three types of estimation.

In the rest of the module we will not do full Bayesian estimation. But we will still use Bayes rule, and take Bayesian ideas in combination with frequentist ideas.

Probabilistic modelling in more general

- We assumed some model generated the data. In this case its was a Bernoulli distribution, with some unknown true parameter value of θ.
- Under the assumption of the Bernoulli model, we used the data to estimate its parameter, and obtain some $\hat{\theta}$.
- Depending on the estimation method we use, our estimates may differ.

- In linear regression, we assume the data is generated by a (normal) distribution centered on some unknown true regression plane; we use the data to estimate it.
- Same principles apply to classification and other tasks.
- Probabilistic modelling is a widely applicable methodology.

Assumptions

fixed, unknown distribution generates the data:

$$y \sim \mathbb{P}\left(y \,|\, x\right)$$

we only see samples, i.e. the training set:

$$\mathfrak{D} = \left\{(x_n, y_n)\right\}_{n=1}^{N}$$

fit model to recover the ground-truth distribution:

$$p(y \,|\, x) \;\approx\; \mathbb{P}\left(y \,|\, x\right)$$

Key Takeaways

- **MLE**: Uses only observed data, can overfit with small samples.
- **MAP**: Incorporates prior, shrinks toward prior belief.
- **Bayesian Mean**: Considers full posterior, integrates over distribution.